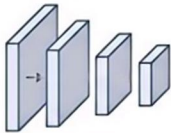
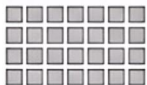


STAGE 0

Pre-trained CNN

 $f_{\theta}(\cdot)$

Calibration Data

 $B = 1024$ samples

STAGE 1

Dual-Branch Scoring

Structural Anchor

 $\perp$
 I_c ∇
 T_c $\mathcal{F}$
 F_c $\rightarrow A_c$

GRA Semantic Branch

Class k_1 GRA $\updownarrow$ Class k_2 $\rightarrow S_c$

STAGE 2

Quality-Gated Refinement

Channel Ranking

keep

bnd

prune

Quality Gate



No

Skip
(anchor only)

Yes

Boundary Swap

kept

pruned

 $\text{Semantic gain} \geq \text{threshold}$

STAGE 3

Structural Pruning

Before pruning



After pruning

Dependency-safe removal

STAGE 4

Fine-tune & Evaluate

Evaluation

Top-1 Acc

Paired Δ

iso-FLOPs

Holm p Hedges' g

Wilcoxon

 $\rightarrow$ Structural path $\rightarrow$ Semantic path (GRA) $\diamond$ Quality gate

Keep / Prune